# CURRICULUM VITAE

## Contact Information

**Ayusha Kumari Sah**
**Inaruwa - 1, Sunsari, Nepal**
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## Personal Details:

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| Name | : Ayusha Kumari Sah |
| Profession | : Geomatics Engineer |
| Date of Birth | : 2058/07/24 B.S (2001/11/09 A.D) |
| Nationality | : Nepali |
| Sex | : Female |
| Permanent Address | : Inaruwa – 1, Sunsari, Nepal |
| Temporary Address | : New Baneshwor, Kathmandu, Nepal |

## Professional Summary:

Licensed Geomatics Engineer with a strong academic foundation and practical experience in GIS, Remote Sensing, surveying, and spatial data analysis. Proficient in utilizing geospatial tools such as QGIS, ArcGIS, and photogrammetry software to generate accurate mapping and analytical outputs. Demonstrates strong analytical thinking, problem-solving ability, and attention to detail, with a commitment to high-quality work. Eager to contribute to a professional organization, gain hands-on experience, and continuously enhance technical expertise while supporting data-driven geospatial and engineering projects.

## Key Skills:

Technical Skills:

- GIS (ArcGIS, QGIS)
- Surveying Techniques
- ENVI
- HEC-RAS
- PostGIS, PostgreSQL
- Parcel Editor
- UAV Mapping
- Web Development
- Efficient in using TS, GNSS/GPS, Plane Table, Theodolite
- Pix4Dmapper
- TBC (Trimble Business Centre)

Soft Skills:

- Communication
- Teamwork
- Problem-solving
- Time Management

## Education:

Bachelor in Geomatics Engineering 2021 - 2026
Kathmandu University, Dhulikhel

Higher Secondary Education Board 2018 - 2020
Greenland International Secondary School, Biratnagar, Morang

Secondary Education 2016 - 2018
Koshi Public Academy, Inaruwa, Sunsari

Lower Secondary Education 2005 - 2016
Koshi Public Academy, Inaruwa, Sunsari

## Professional Experience:

**GIS Intern | Kathmandu Metropolitan City | 26th November – 23rd February**

- Identification and verification of government and public land using cadastral maps and ward-wise cadastral records of Kathmandu Metropolitan City using QGIS.
- Digitization of buildings of various wards of KMC from Orthophoto for House Numbering System in Kathmandu Metropolitan City using QGIS.
- Review of land use planning and house numbering projects of Kathmandu Metropolitan City.

## ACADEMIC & TECHNICAL PROJECTS:

**Assessment Of Rooftop Solar Energy Potential Using UAV-Based Roof Modelling**

— Tools Used: UAV for data acquisition, Ground Control Points (GCPs) with GNSS for accurate georeferencing, photogrammetry software (Pix4D) for 3D roof modelling and DSM generation, and GIS tools for estimating rooftop solar energy potential.

**Integrated Approach to Waste Management in Banepa Municipality: Combining Geographic Information System, Analytical Hierarchy Process, and Network Analysis for Landfill Site Selection and Route Optimization**

—Tools Used: GIS, AHP, and Network Analysis, leveraging Handheld GPS, ArcGIS, and QGIS.

**3D Model Of KU Cafe and Amphitheater**

—Tools Used: Measuring tools and scale reduction

**Mini Project: Selection Of Large Scale Solar Photovoltaic Project Using AHP Method in Sunsari District**

—Tools Used: AHP and GIS

**Mini Project: Image Processing To Make Topographical Map**

—Tools Used: Pix4D Mapper and ArcGIS

**Mini Project: Determining Shortest Route From Inaruwa To Kathmandu University Using pgRouting**

—Tools Used: pgRouting with PostGIS

**Mini Project : Thematic Map**

—Tools Used: ArcGIS

**Mini Project : Land Use Land Cover Map**

—Tools Used: QGIS

**Mini Project : Flood Modeling of the Seti River in Shuklagandaki Municipality**

—Tools Used: HEC-RAS and GIS Analysis

**Challenges and Opportunities in Addressing Informal Settlements in Urban Nepal: A Case Study of Kathmandu Valley**

**Comparative Analysis of Project Evaluation Practices: A Cross-Country Perspective**

## CERTIFICATIONS & TRAININGS:

- QGIS Training Session emphasized on Use of OSM Data and their Applications in Attaining Sustainable Development Goals
  - Mapping Banepa Municipality in OSM and it's Application in Mitigating Effects of Urbanization- A case study of Urban Rooftop Farming
- ArcGIS Story Map Training
- Workshop on Navigating Future of AI and Web GIS in Geomatics

## PUBLICATIONS:

### Published in FIG (International Federation of Surveyors)

Integrated Approach to Waste Management in Banepa Municipality: Combining Geographic Information System, Analytical Hierarchy Process, and Network Analysis for Landfill Site Selection and Route Optimization

https://www.fig.net/resources/proceedings/fig_proceedings/nepal/papers/ts01a/TS01A_rijal_timalsina_et_al_12908.pdf

## ACHIEVEMENTS:

NEB Scholarship - +2 Science (Extra Math)

35% Scholarship - Bachelor in Geomatics Engineering

## EXTRACURRICULAR ACTIVITIES:

- General Member at Geomatics Engineering Society(GES)
- Active participant in academic seminars and workshops
- Volunteer in Annual Event of Geomatics Engineering Society

## LANGUAGES:

Maithili - Native
Nepali – Fluent
English – Fluent
Hindi - Fluent